

Homework 3

“Vue.js and Google Cloud Firestore”

The following problem set is worth 100 points total. Please submit a zip file containing all relevant code to the Canvas drop box by the due date of the assignment.

Requirements

In this course, we are going to look at a series of JavaScript frameworks for developing applications. The first of the frameworks is Vue.js. This assignment will also familiarize you with creating a Google Firestore database.

DDR (100 pts)

For the third homework assignment, you are going to implement a playable game of **DDR**:



This game is based off the music video game series produced by Konami called **Dance Dance Revolution (DDR)**. DDR was popularized at arcades and then later found its way to consoles, both home (PlayStation) and portable (Nintendo DS).



Idea of the game

For an idea of what the game is like, a playable version can be found here:

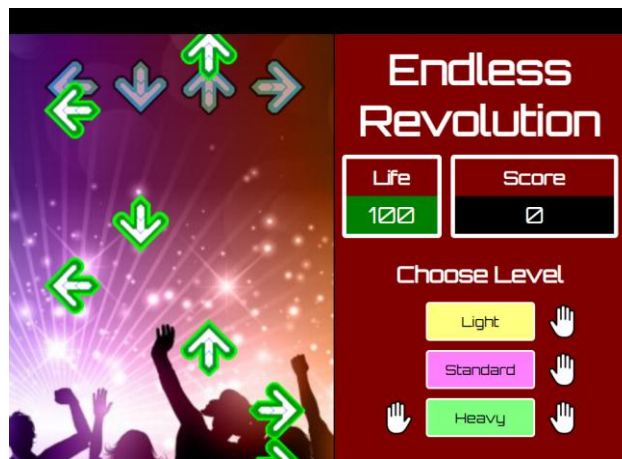
<https://eileenho.github.io/keyboard-DDR/index.html>

In the classic arcade game, players are presented with a series of arrows (up, down, left, right) and use their footwork to match the right arrow on the screen. With this game you are to implement the same thing, however you will use the **arrow keys on your keyboard**.

Assignment Requirements

You will need to implement a **DDR-type** game using Vue.js. For handling the four arrow keys, you can use the Vue.js key press events.

Drawing the game board



Building the game is relatively simple. In the screenshot above, arrows flow from bottom to top. You can use the **setInterval** function on the window object to keep adding rows. That is (as an example) each second you can randomly pick one of the four arrows and that will be the new bottom row. All other rows will be pushed up by one.

Additional Requirements

- Give the user several incorrect moves they are allowed to make (e.g. a health bar)
- Give the user the choice of game (light, medium, hard), which dictates how quickly the **setInterval** function operates
- Each correct move you'll add some value to their score
- When the user runs out of incorrect moves, the game is over
- When the game is over, ask the user if they would like to play again
- Store the high scores in a Firebase database as well (showing the users with the current high score)

Submission Requirements

For submission, upload a zip file containing all assets I need to run your Vue.js game. If you are using Node.js for your Vue implementation, do not include the **node_modules** directory in your submission.

Melusky – CMPSC421 – Spring 2024

You do not need to use Node for this assignment (I'll go over Vue in the coming weeks in the course).
Submit using the Canvas drop box before the due date.